Home Lab 12: The Soil Jar (Settling) Test

Do this with a parent. Wash your hands after handling soil. Clean up after.

Goal

Separate a soil sample into sand, silt, and clay layers and estimate its texture class.

You need

From home: 2 clear jars with tight lids (jam jars work). A trowel or big spoon. Water. A drop of dish soap. A ruler. Soil from **two different spots** — e.g. your yard and a park, or a garden bed and a bare path. Label the jars.

Steps

1. Fill each jar about ⅓ with soil (break up clumps, pick out rocks, twigs, roots).
2. Add water until the jar is nearly full. Add **one drop of dish soap** (helps the clay separate).
3. Lid on tight. **Shake hard for 2 minutes.**
4. Set the jar somewhere it won't be bumped. Watch it settle:
    - **Sand** drops out in the first minute (bottom layer).
    - **Silt** settles over a few hours (middle layer).
    - **Clay** takes a day or more; the water may still look cloudy.
5. After a day, measure each layer's thickness with the ruler.

Results

	Jar 1 (____)	Jar 2 (____)
Sand layer (mm)		
Silt layer (mm)		
Clay layer (mm)		
Total (mm)		
% sand		
% silt		
% clay		
Texture class (from the triangle)		
Soil color		

(% = layer thickness ÷ total thickness × 100)

Follow-up

1. Which of your two soils had more sand? Which had more clay? Which would drain water faster?
2. Use a soil texture triangle to name each soil's class.
3. What does the **color** of each soil tell you (red = iron oxides; dark = organic matter; gray = waterlogged; pale = lime/salt/leached)?
4. If soil found on a suspect's boots was 70% sand, 20% silt, 10% clay, which of your two soils would it be more likely to have come from?